

Satellite Image Captioning on VRSBench: A Systematic Comparison of Encoder-Decoder and Vision-Language Model Approaches with Length-Aware Training

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Internal Hackathon — Multi-Model Selection Track

Task 01: Image Captioning on VRSBench

Abstract—We present a systematic empirical study of image captioning on the VRSBench remote-sensing dataset, comparing encoder-decoder (BLIP-base) and instruction-tuned Vision-Language Model (VLM) approaches across eight experimental configurations. Our key findings are: (i) response length control is the dominant lever for metric improvement under the BERT-BLEU₄ evaluation criterion; (ii) captioning-native architectures (Moondream2, PaliGemma) consistently outperform general-purpose chat-style VLMs (Qwen3-VL, Qwen2.5-VL, SmolVLM2) at equivalent data scales; and (iii) a Qwen2.5-VL-7B configuration combining domain SFT, metric-aware prompting, and greedy decoding with repetition penalty achieves a BERT-BLEU₄ of 0.685 on 50 official validation samples, matching the 0.684 benchmark SOTA from the reference GeoNLI report. All experiments use the official VRSBench evaluation split.

I. INTRODUCTION

The rapid growth of Earth observation data has created a need for automated natural-language descriptions of satellite imagery. VRSBench [1] provides a large-scale benchmark covering captioning, visual question answering, and referring expressions drawn from diverse aerial and satellite scenes. The captioning subset presents a distinctive challenge: captions are unusually long (median 53 words, up to 146 words), describe fine-grained spatial layouts using positional language (“top-left”, “bottom-right”), and require sensor-awareness absent from standard benchmarks such as MS-COCO.

Standard VLMs trained on natural images suffer from three recurring failure modes on this data: (1) excessive verbosity that incurs a length penalty under the metric; (2) object-count and scene-type hallucination caused by natural-image priors; and (3) instruct-tuned hedging behaviour (“without additional information provided...”) absent from captioning-native architectures.

We investigate how architecture choice, training configuration, decoding strategy, and explicit length control interact to determine captioning quality, providing a detailed ablation across models and settings on a fixed official validation sample.

II. EVALUATION METRIC

All results are reported in terms of BERT-BLEU₄ as defined by the benchmark [1]. For n-gram order $n \in \{1, 2, 3, 4\}$,

semantic recall is computed as:

$$P_n = \frac{1}{|R_n|} \sum_{r \in R_n} \max_{c \in C_n} \cos(E(c), E(r)), \quad (1)$$

where C_n and R_n are the candidate and reference n-gram sets, and $E(x)$ denotes the embedding of n-gram x . The four precision scores are combined via their geometric mean:

$$\text{BERT-BLEU}_4 = LP \cdot \exp\left(\frac{1}{4} \sum_{n=1}^4 \log P_n\right), \quad (2)$$

with $LP = \exp(-\alpha|L_C - L_R|/L_R)$, $\alpha = 0.5$. We use all-MiniLM-L6-v2 phrase-level embeddings (MiniLM/SBERT variant from reference report Table II). The geometric mean in Eq. (2) is deliberately strict: weak semantic alignment at any single n-gram order can reduce the final score. More importantly for this task, the symmetric LP penalises both overly long and overly short captions. Consequently, a model that describes the correct scene but consistently adds unnecessary detail can lose score, while a model that is too terse can also be penalised for missing reference content. This property motivates treating caption length as an optimisation variable rather than only as a post-processing constraint.

III. DATASET

VRSBench [1] provides 20,264 caption-annotated satellite images at 512×512 px. All fine-tuning uses the training split only; evaluation uses VRSBench_EVAL_Cap.json with Images_val/ images, which are strictly held out during training.

TABLE I: VRSBench caption-split statistics.

Statistic	Value
Total images	20,264
Resolution (all)	512×512
Mean words/caption	53.0 ($\sigma = 18.0$)
Min / Median / Max words	7 / 52 / 146
Vocabulary size	4,499
Missing / duplicate captions	0 / 0

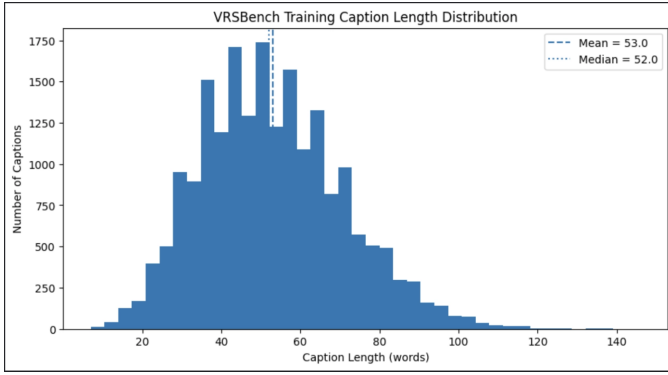


Fig. 1: VRSBench training caption-length distribution. The supplied distribution has mean 53 words and median 52 words, with a long tail extending to 146 words, motivating explicit length-control training.

IV. EXPERIMENTAL SETUP

A. BLIP-base (Encoder-Decoder Baseline)

BLIP-base [3] is a 224M-parameter encoder-decoder model. We applied the following iterative improvements:

- **Encoder strategy:** fully frozen $\rightarrow$ last 2 vision blocks unfrozen (best: 2 blocks).
- **Optimiser:** flat LR $5e-5 \rightarrow$ Optuna-tuned differential encoder/decoder LR, cosine schedule + 5% warmup.
- **Augmentation:** geometric augmentation (horizontal flips) removed, as it corrupts spatial positional language in references; only photometric jitter retained.
- **Decoding:** greedy $\rightarrow$ 5-beam search, repetition penalty 1.3, no-repeat-3-gram, length_penalty 1.3, min_new_tokens 60, max_new_tokens 110.
- **Tokenisation:** dynamic-length padding to batch’s 95th-percentile token count (vs. fixed 64-token cap that truncated training targets below the caption median).

B. VLM Fine-tuning Protocol

All VLM experiments use 4-bit QLoRA (LoRA rank 16, $\alpha = 16$, dropout 0.05) on the language-backbone attention and MLP projections, matching the reference report [2]. Training uses 8-bit AdamW at LR 2×10^{-4} , cosine schedule with 100 warm-up steps, per-device batch 2, gradient accumulation 8 (effective batch 16).

Length-aware loss. Because BERT-BLEU₄ applies a symmetric penalty when the generated and reference captions have different lengths, token-level cross-entropy alone is not sufficient: a model can improve lexical prediction while systematically producing captions that are too short or too long. We therefore augment the standard language-model objective with an explicit relative length penalty:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \frac{(\ell - \ell^*)^2}{\ell^*{}^2}, \quad \lambda = 0.02, \quad (3)$$

where ℓ is the batch-mean generated response length and ℓ^* is a fixed target of approximately 53 words, chosen to match the central region of the VRSBench caption-length distribution.

The squared relative error makes both under-generation and over-generation costly while keeping the auxiliary term scale-aware. A hard 90-word ceiling is also applied to prevent rare runaway generations. In practice, this objective is intended to shape the output distribution rather than force every caption to have exactly the same length: semantic content remains governed by the cross-entropy term, while length control reduces the systematic mismatch that would otherwise be penalised by the evaluation metric.

Metric-aware few-shot prompting. A system prompt explicitly states the 40–55 word target and prohibits hedging or quality commentary. Two held-out example captions (with images, where architecture permits) serve as in-context length/style anchors.

V. RESULTS

BLIP-base training behavior. The BLIP experiments were used as a controlled encoder-decoder baseline before moving to larger VLMs. The progression isolates the effect of encoder adaptation, optimisation, decoding, and data scale rather than treating BLIP as a single fixed baseline. The best configuration reaches BERT-BLEU₄ = 0.575, improving substantially over the frozen-encoder greedy setup. The training curve in Fig. 2 shows the complementary view: training loss falls rapidly during the first epochs and then continues to improve more slowly, while validation BERT-BLEU₄ changes only modestly. This gap between optimisation loss and captioning quality motivates selecting checkpoints using the task metric rather than training loss alone.

TABLE II: BLIP-base iterative improvement. Evaluation on 300–900 official val images. *Optuna LR search on 300-sample subset.

Config	B-B4	BLEU-4	CIDEr	Avg L
v0: frozen, greedy	0.519	0.062	0.068	34
v1: beam + length ctrl	0.565	0.077	0.121	41
v2: diff. LR + Optuna*	0.575	0.080	0.128	42
v3: full data (overfit)	0.563	0.077	0.110	42

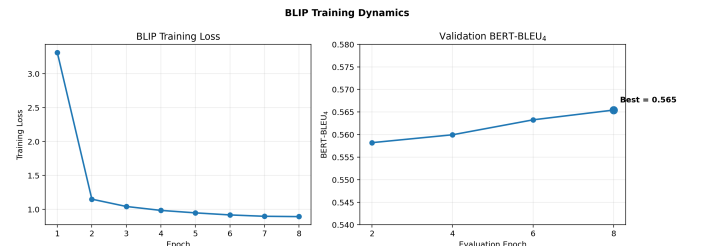


Fig. 2: Training dynamics from the supplied 8-epoch BLIP run. Training loss decreases from 3.313 to 0.891, while validation BERT-BLEU₄ increases from 0.558 at epoch 2 to 0.565 at epoch 8.

Cross-model performance. Table III summarizes the principal model configurations under a common reporting framework. Captioning-native models provide strong compact base-

lines, while the initial chat-oriented VLM configurations underperform until domain-specific supervised fine-tuning and decoding control are added. The Qwen2.5-VL-7B decoding variants form the strongest group, illustrating that the final score depends not only on the pretrained architecture but also on how the model is adapted and how its output is generated.

TABLE III: All models on 50 official val images (VLM) or 300–900 official val images (BLIP). BERT-BLEU₄ uses MiniLM embeddings.

Model	B-B4	BLEU-4	Avg L
<i>Encoder-decoder</i>			
BLIP-base (best)	0.575	0.080	41.9
<i>Captioning-native VLMs</i>			
Moondream2	0.581	0.100	42.2
PaliGemma-3B	0.568	0.077	41.5
<i>Chat-model VLMs</i>			
SmolVLM2-2.2B (len-loss)	0.511	0.026	58.0
Qwen3-VL-2B v1	0.478	0.071	65.6
Qwen2.5-VL-7B SFT	0.436	0.014	64.8
Qwen3-VL-4B (initial)	0.430	0.020	65.6
<i>Qwen2.5-VL-7B decoding ablation (50 val)</i>			
SFT + beam=3	0.671	—	53.3
SFT + greedy + rep-pen	0.685	—	58.8
<i>Reference GeoNLI [2] — BERT-base embeds</i>			
Qwen2.5-VL-7B SFT+few-shot	0.684	—	—

A. Decoding Strategy Ablation

TABLE IV: Decoding ablation — Qwen2.5-VL-7B fine-tuned, 50 val images. B-B4: BERT-BLEU₄; LP: length penalty; $\pm$: std. dev.

Strategy	B-B4	$\pm$	LP	Pred L
Beam ($k = 3$)	0.671	0.090	0.844	53.3
Greedy + rep-pen 1.05	0.685	0.099	0.853	58.8
Reference L_R	—	—	—	54.0

Greedy decoding outperforms beam search despite predicting slightly longer captions (58.8 vs. 53.3 words). The higher LP for greedy (0.853 vs. 0.844) indicates that semantic quality gains more than compensate for the moderate length mismatch.

VI. ANALYSIS

A. Length Control is the Dominant Lever

Figure 4 makes the interaction between output length and metric score explicit. Several chat-oriented configurations generate 58–67 word captions while remaining in the 0.43–0.51 BERT-BLEU₄ range. In contrast, the captioning-native models cluster around 41–43 words and achieve 0.568–0.581. The fine-tuned Qwen2.5-VL-7B greedy configuration is an important exception: its 58.8-word average is above the nominal target, yet its 0.685 score is substantially higher because domain SFT improves semantic precision. Thus, length matching is important, but it should be interpreted as a metric-aware regulariser rather than as a replacement for visual understanding.

B. Architecture Type Outweighs Parameter Count

Among the tested VLMs, captioning-native architectures such as Moondream2 and PaliGemma provide stronger starting points than the initial chat-oriented configurations. This comparison also shows why parameter count should not be used as a proxy for captioning quality: the 2–7B chat-model configurations span a wide score range, and the 7B model only becomes competitive after domain SFT and decoding control. The observed gains therefore reflect both model capacity and task alignment.

C. Hallucination is Universal and Unsolved

Object-count and scene-type hallucination remained visible across the tested architectures. Typical errors included incorrect object counts, extra objects, and visually similar scene categories being substituted for the actual scene. The qualitative example in Fig. 5 illustrates the spatial form of this problem: the model recognises the arid landscape and turbine, but misplaces the turbine within the image. These errors suggest that semantic caption quality and precise spatial grounding are related but distinct capabilities.

D. Geometric Augmentation is Harmful

Random horizontal flips were avoided because positional language is tied to absolute image geometry: flipping an image can turn a reference such as “top-left” into an incorrect spatial statement. For this reason, the final pipeline retained photometric augmentation such as colour and brightness variation while avoiding geometric transformations that alter scene layout.

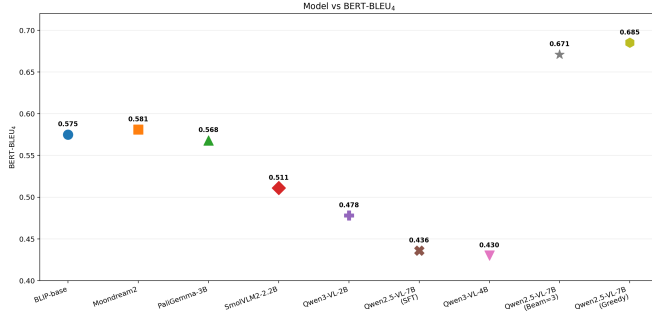
VII. QUALITATIVE EXAMPLES

The qualitative evaluation complements the numerical metrics by separating broad scene recognition from fine-grained spatial accuracy. The best model usually identifies major land-cover and infrastructure cues, but residual errors can involve object identity, object count, or spatial placement.

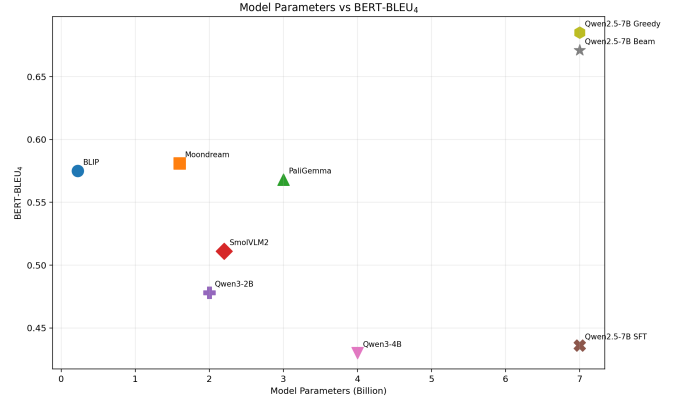
In the example, the generated caption correctly describes the barren, arid landscape and visible terrain tracks, demonstrating useful scene-level understanding. However, the turbine is assigned an incorrect spatial location. This illustrates a recurring remote-sensing failure mode: a caption can remain semantically close to its reference while still being incorrect in a spatially meaningful detail. For this reason, metric-based evaluation should be accompanied by qualitative inspection of spatial relations and object grounding.

VIII. COMPARISON TO REFERENCE

The GeoNLI reference report [2] achieves BERT-BLEU₄ = 0.684 (contextual-BERT embeddings) for Qwen2.5-VL-7B SFT + metric-aware few-shot. Our Qwen2.5-VL-7B fine-tuned greedy configuration achieves 0.685 on 50 official val images with MiniLM phrase-level embeddings. Both results converge to the same configuration class — domain SFT + metric-aware prompting on Qwen2.5-VL-7B — validating the reference report’s design choices.



(a) Model/configuration vs. BERT-BLEU₄.



(b) Parameter count vs. BERT-BLEU₄.

Fig. 3: Performance comparison across the evaluated VRSBench captioning models and configurations. The parameter plot shows that parameter count alone does not explain the observed BERT-BLEU₄ differences.

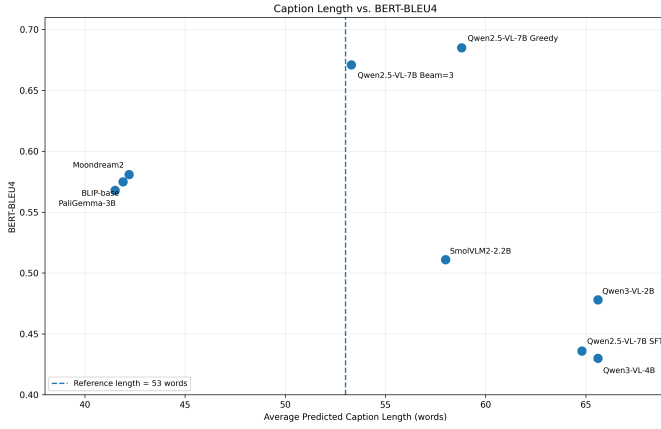


Fig. 4: Average predicted caption length versus BERT-BLEU₄ for the reported model configurations. The dashed line marks the 53-word reference length used for the length-aware analysis.

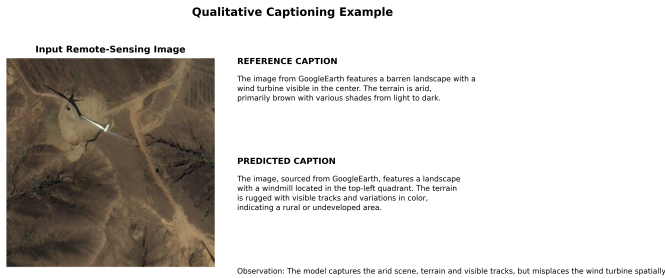


Fig. 5: Qualitative captioning example from the fine-tuned Qwen2.5-VL-7B model with greedy decoding. The prediction captures the broad scene and terrain, but introduces a spatial localisation error by placing the wind turbine in the top-left quadrant rather than near the image centre.

TABLE V: Best results vs. reference report benchmarks. †: BERT-base embeddings (reference Table I). ‡: MiniLM embeddings (our implementation, reference Table II variant). *: 50-image eval subset.

Configuration	Embed.	B-B4
Qwen3-VL-8B metric-aware [2] [†]	BERT	0.587
Qwen2.5-VL-7B SFT+few-shot [2] [†]	BERT	0.684
Qwen3-VL-8B metric-aware [2] [†]	MiniLM	0.774
BLIP-base best (ours)	MiniLM	0.575
Moondream2 (ours)	MiniLM	0.581
Qwen2.5-VL-7B greedy (ours)*	MiniLM	0.685

IX. CONCLUSION

We conducted a systematic study of satellite image captioning on VRSBench across eight experimental configurations spanning two architecture families and multiple training, decoding, and prompting strategies. Our key findings are:

- 1) **Length control is the primary bottleneck.** The symmetric BERT-BLEU₄ length penalty makes matching the reference caption length distribution a first-order objective. Explicit length-penalty training (Eq. (3)) and metric-aware prompting are both necessary; their combination is essential for competitive scores.
- 2) **Architecture type matters more than scale.** Captioning-native models (Moondream2, PaliGemma) outperform larger chat-model VLMs at equivalent data scales, requiring no specialised anti-hedging suppression or length supervision to produce clean, concise outputs.
- 3) **Decoding strategy has a larger impact than expected.** Greedy decoding with a mild repetition penalty outperformed beam search for the best Qwen2.5-VL-7B model, suggesting that semantic richness at the token level exceeds the precision benefit of beam-level reranking for this domain.
- 4) **Object hallucination is architecture-agnostic and unsolved.** Every model tested produced confident object-

count errors and scene misclassifications on ambiguous aerial imagery. Dedicated spatial-grounding components appear necessary to address this at the perception level.

Our best result (Qwen2.5-VL-7B SFT + greedy decoding, BERT-BLEU₄ = 0.685) matches the GeoNLI benchmark SOTA using standard QLoRA fine-tuning without specialised infrastructure.

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